

PREDICTING ONLINE SHOPPERS' PURCHASING INTENTION

COURSE: MIS 451

LECTURER: HUYNH GIA LINH

QUARTER 3, 2025-2026

PRESENTED BY : POLYGON



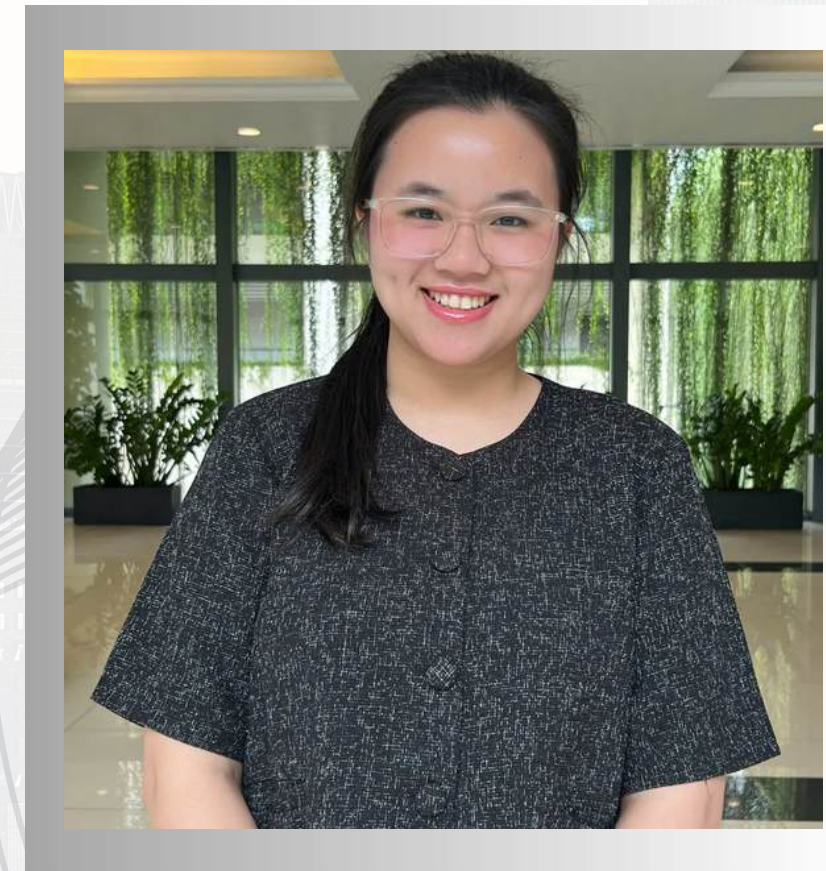
OUR TEAM



Nguyen Thi Hoai Thuong
2232300040



Tran Ngoc Trung Hieu
2132300543



Nguyen Ngoc Minh Thu
2232300163

TABLE OF CONTENT

- 1. Business Context and Objectives**
- 2. Data collection**
- 3. Exploratory Data Analysis**
- 4. Data Cleaning and Transformation**
- 5. Model Development and Evaluation**
- 6. Interpretation and Insights**

1.

BUSINESS CONTEXT

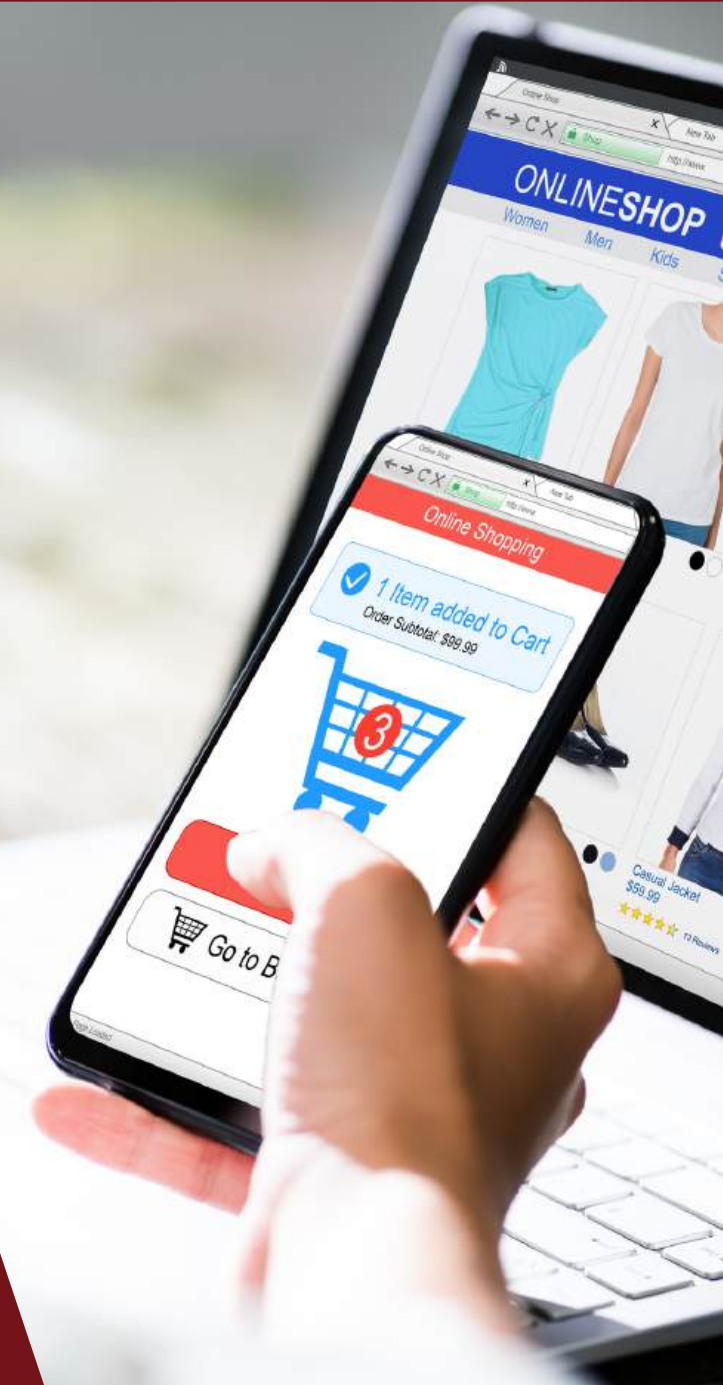
and OBJECTIVES



BACKGROUND

Predicting whether an online visitor will make a purchase is a key challenge in e-commerce. Session-level behavioral data (e.g., pages viewed, time spent, bounce rate) can provide strong signals of purchase intention. This study uses the UCI Online Shoppers dataset.

- **Dataset:** 12,330 sessions with revenue as the target variable
- **Problem:** Imbalanced data (few purchases vs many non-purchases)
- **Business value:** Enables targeted marketing and personalized interventions
- **Academic support:** Prior studies show clickstream data predicts buying behavior
- **Goal:** Build ML models and identify key factors influencing purchase intention



1.2 RESEARCH QUESTION



Which machine learning classification model best predicts whether an online shopping session will result in a purchase?

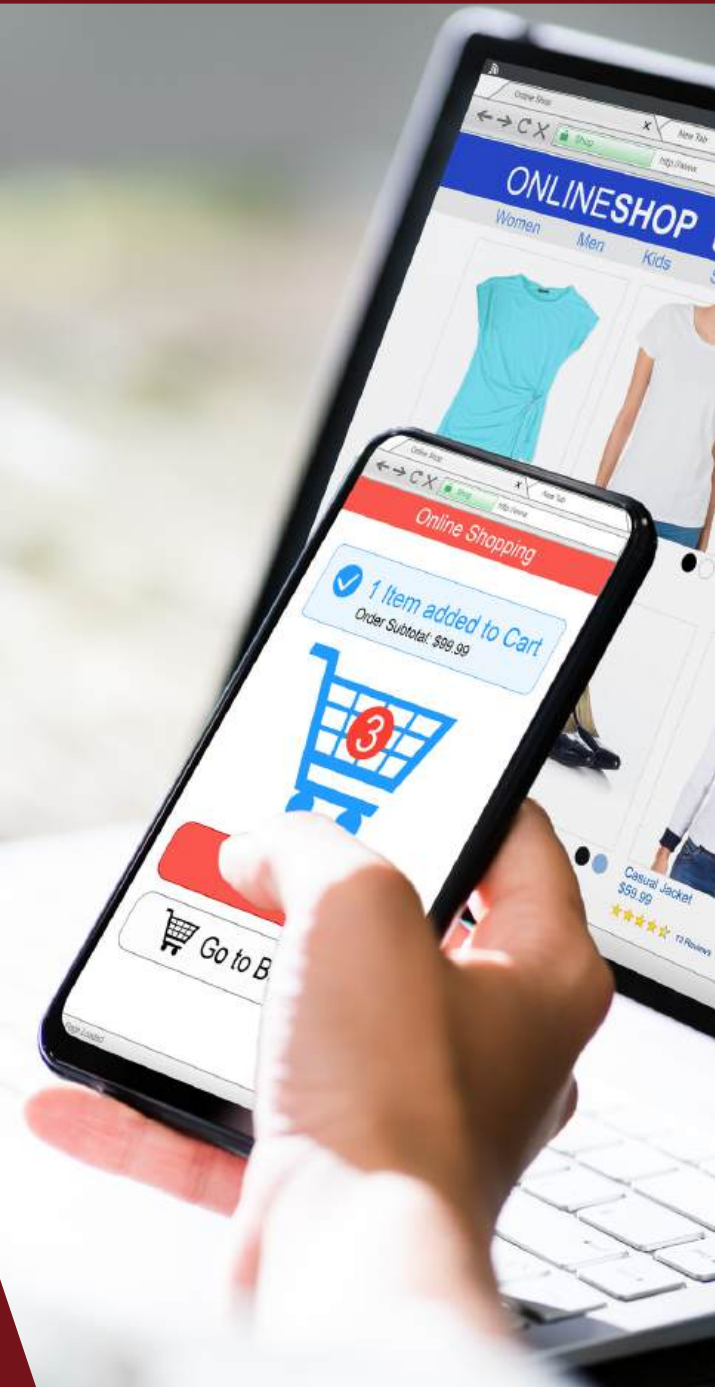


1.3

OBJECTIVES

Develop a machine learning model to **predict whether an online session will generate revenue.**

- Analyze **session behavior** (pages, duration, bounce/exit rate, etc.)
- **Compare models:** Logistic Regression, KNN, Random Forest, MLP
- Use **F1-Macro** due to imbalanced data
- Identify **key features influencing purchase intention**
- Provide **insights for marketing and conversion improvement**



2.

DATA COLLECTION

**DESCRIBE THE DATASET, ITS SOURCE,
AND KEY CHARACTERISTICS**



2.1 DATASET SOURCE AND STRUCTURE

- **Source:** Online Shoppers Purchasing Intention Dataset (UCI Repository)
- **Size:** 12,330 sessions × 18 features
- **Period:** February–December (excluding January and April)
- **Unit of Analysis:** Each row represents one unique user session
- **Data Design:** Sessions belong to different users, reducing bias from repeat visits
- **Limitation:** Seasonal patterns may not fully represent a complete calendar year

Data Quality Check

- **Missing Values:** None detected
- **Duplicate Records:** 125 rows identified and removed
- **Final Dataset Size:** 12,205 rows × 18 features



2.2 FEATURE DESCRIPTION

Feature	Type	Description
Administrative	Numeric	Number of administrative pages visited
Administrative_Duration	Numeric	Time spent on administrative pages (seconds)
Informational	Numeric	Number of informational pages visited
Informational_Duration	Numeric	Time spent on informational pages (seconds)
ProductRelated	Numeric	Number of product-related pages visited
ProductRelated_Duration	Numeric	Time spent on product-related pages (seconds)
BounceRates	Numeric	Percentage of sessions with no interaction
ExitRates	Numeric	Percentage of pageviews that ended the session
PageValues	Numeric	Estimated value of pages visited before purchase
SpecialDay	Numeric	Closeness to a special day (0–1)

2.2 FEATURE DESCRIPTION

Feature	Type	Description
Month	Categorical	Month of the session
OperatingSystems	Numeric	Visitor's operating system (encoded)
Browser	Numeric	Visitor's browser (encoded)
Region	Numeric	Geographic region (encoded)
TrafficType	Numeric	Traffic source (encoded)
VisitorType	Categorical	Visitor type (Returning, New, Other)
Weekend	Boolean	Session occurred on a weekend
Revenue	Boolean (Target)	Whether the session resulted in a purchase

3.

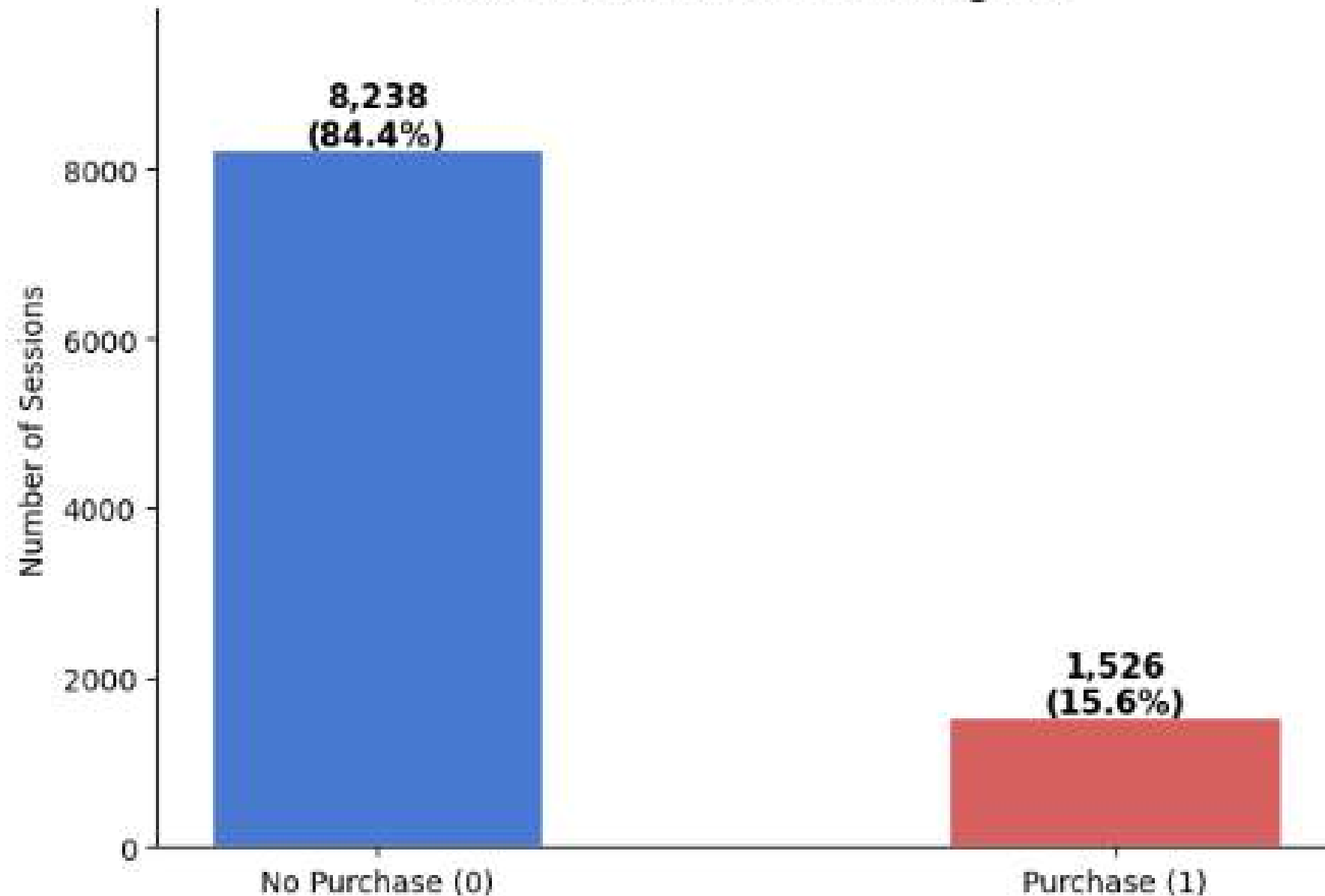
EXPLORATORY DATA ANALYSIS PRESENT THE MAIN FINDINGS AND INSIGHTS FROM DATA EXPLORATION

All EDA was conducted on the **training set** only (80% of the data) to **prevent data leakage** from the test set.

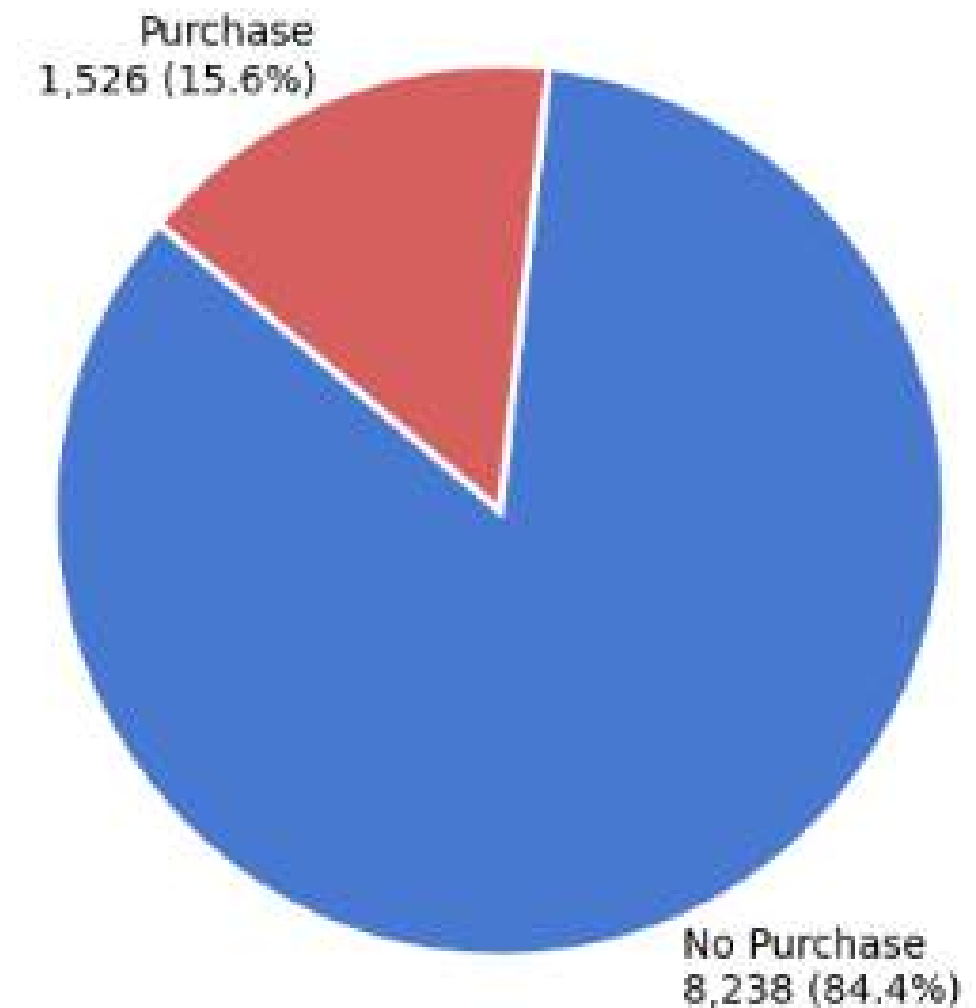


CLASS IMBALANCE: THE CENTRAL CHALLENGE

Class Distribution — Training Set



Purchase vs No-Purchase Ratio



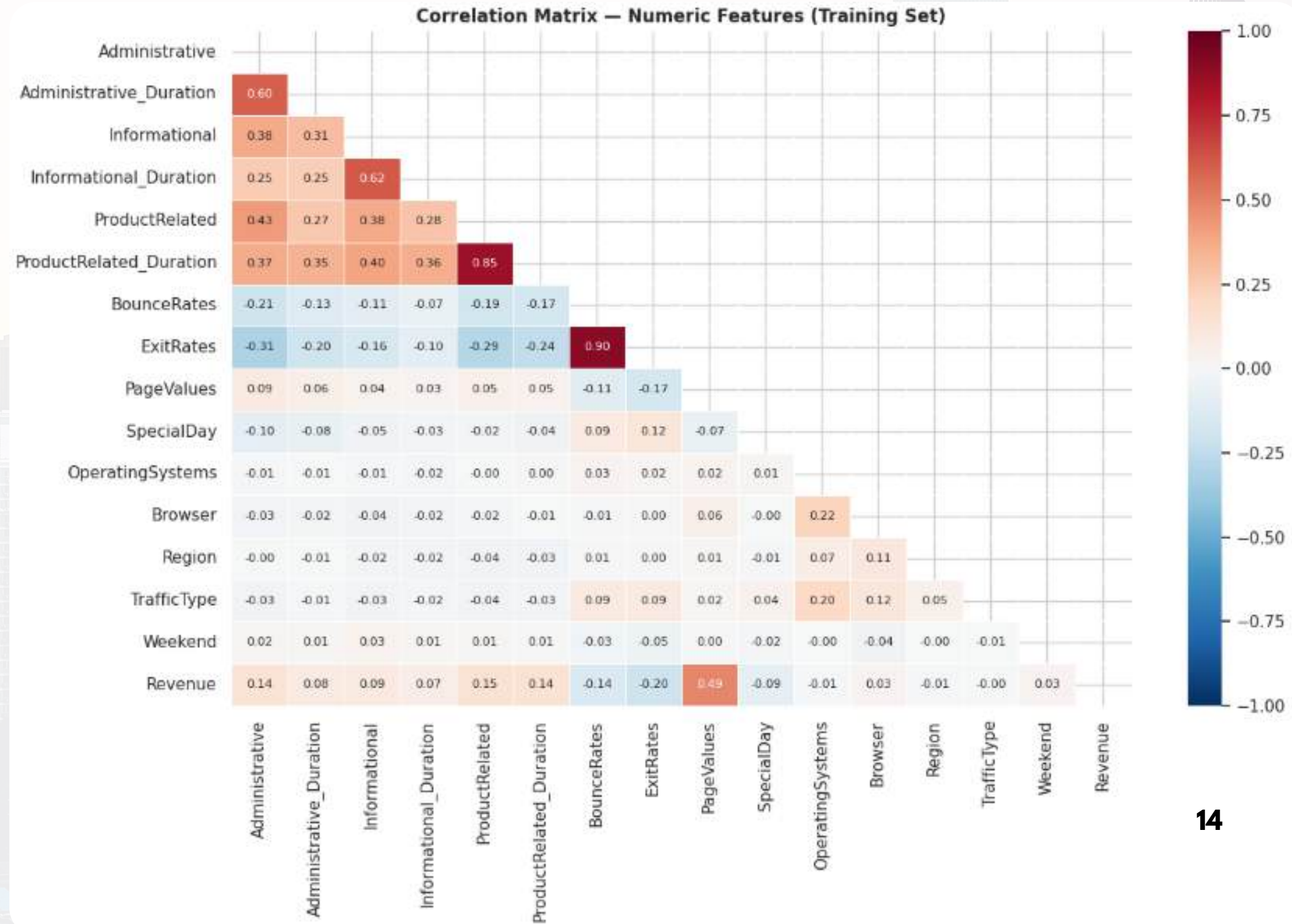
Target Variable Distribution in Training Data

- Only **15.6% of sessions lead to purchases (~1:5 ratio)**, creating strong class imbalance.
- A naive model predicting **“no purchase” already achieves 84.4% accuracy**, making accuracy¹³ unreliable. Therefore, class balancing and Macro F1 are used for evaluation.

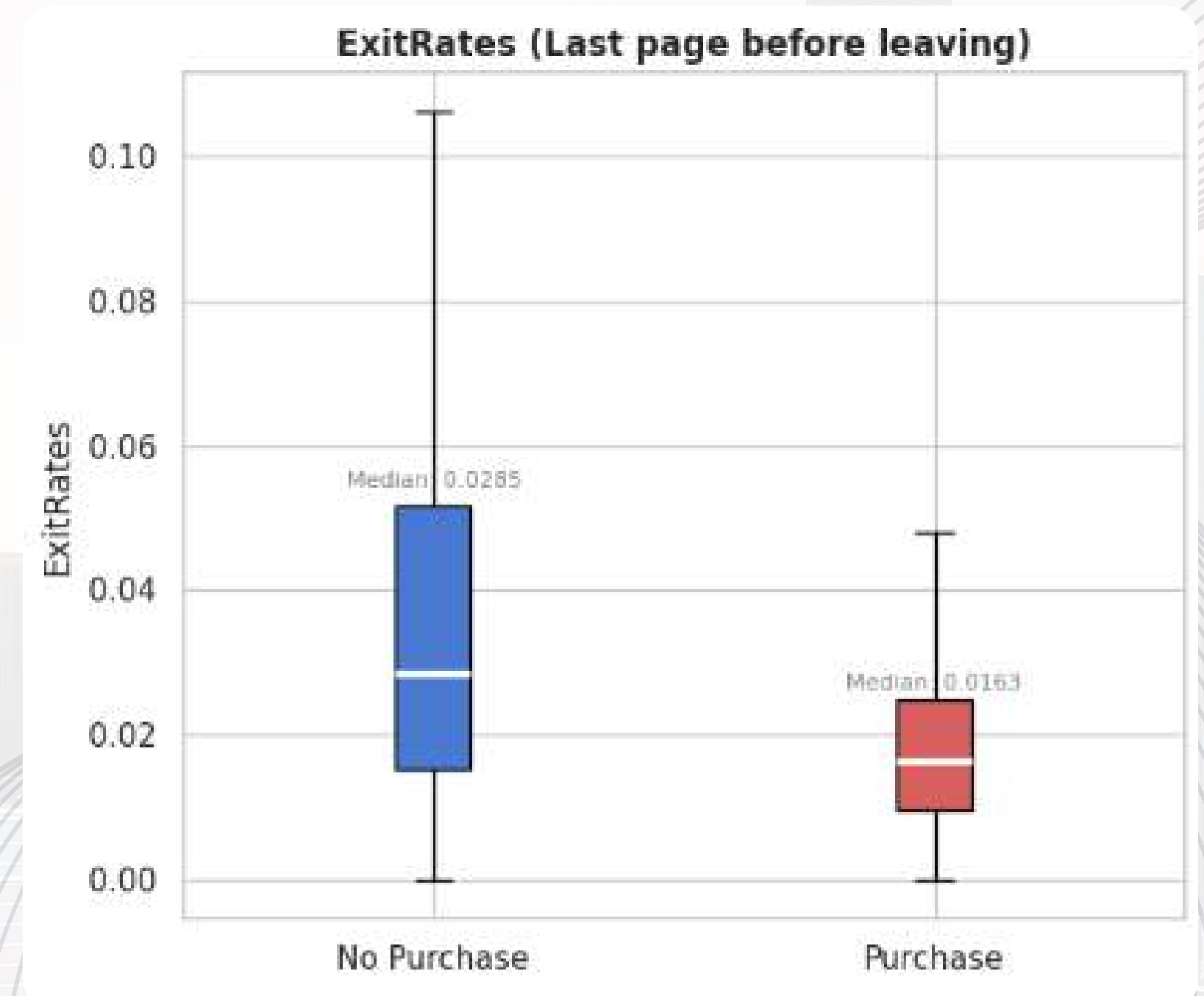
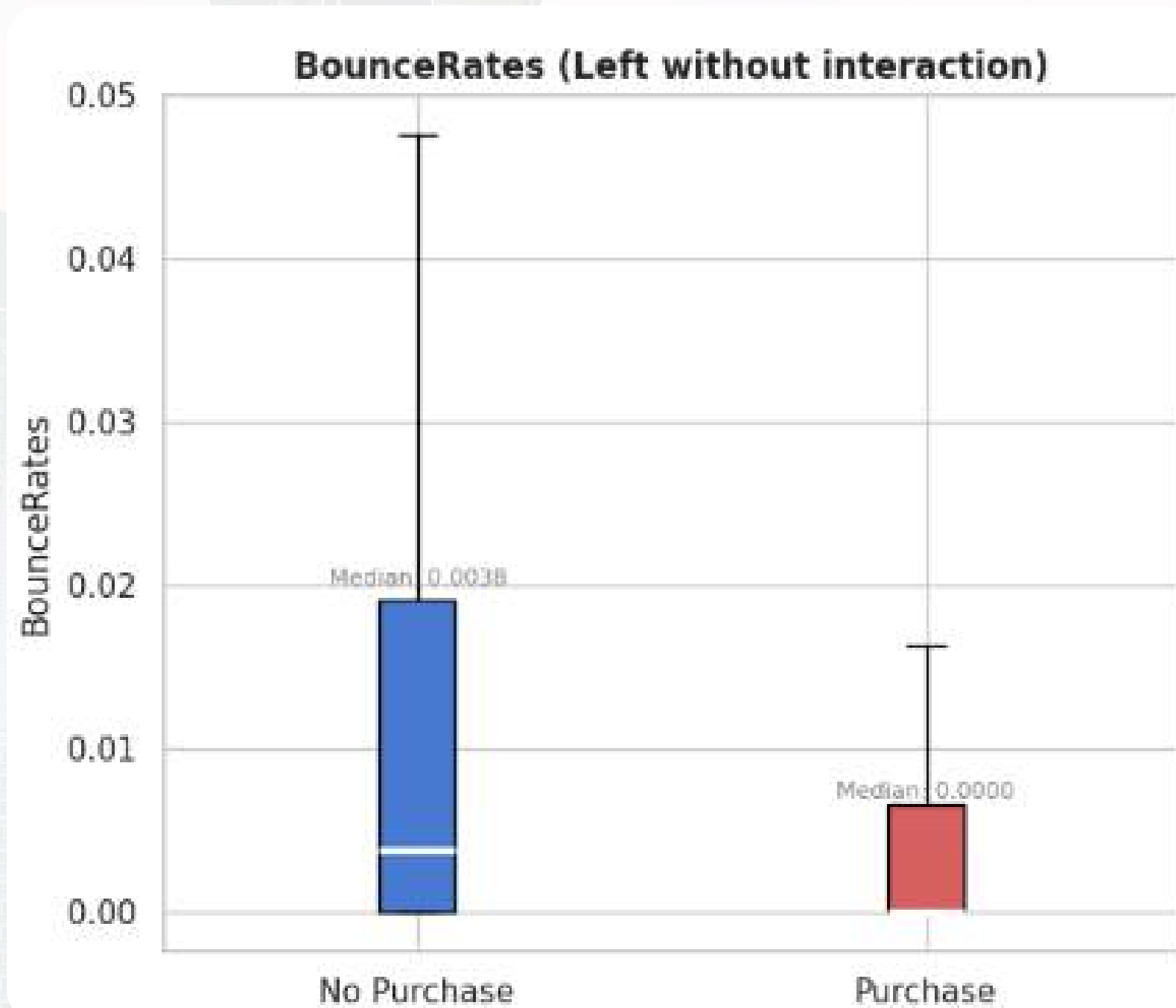
3.2

PAGEVALUES DOMINATES ALL OTHER FEATURES

- PageValues has **the strongest** link to Revenue (+0.49)
- Indicates proximity to checkout (high purchase intent)
- Available in real time → **actionable for interventions**
- Other features are weaker individually
- The model needs **feature combinations** for better prediction



DISENGAGEMENT BEHAVIOR AND MULTICOLLINEARITY

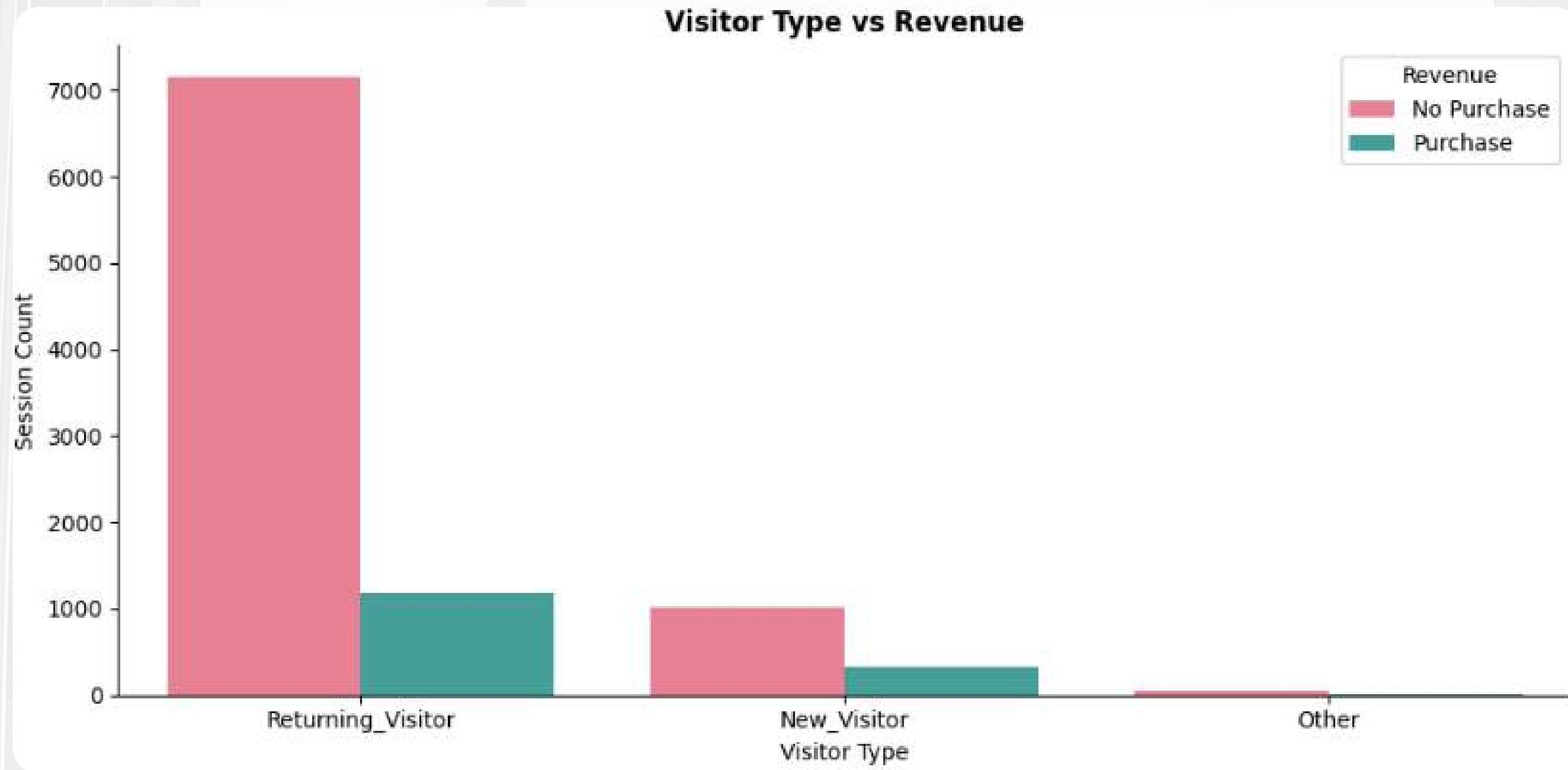


Distribution of Exit Rates by Purchase Outcome

Distribution of Bounce Rates by Purchase Outcome

- Only **15.6% of sessions lead to purchases (~1:5 ratio)**, creating strong class imbalance.
- A naive model predicting **“no purchase” already achieves 84.4% accuracy**, making accuracy¹⁵ unreliable. Therefore, class balancing and Macro F1 are used for evaluation.

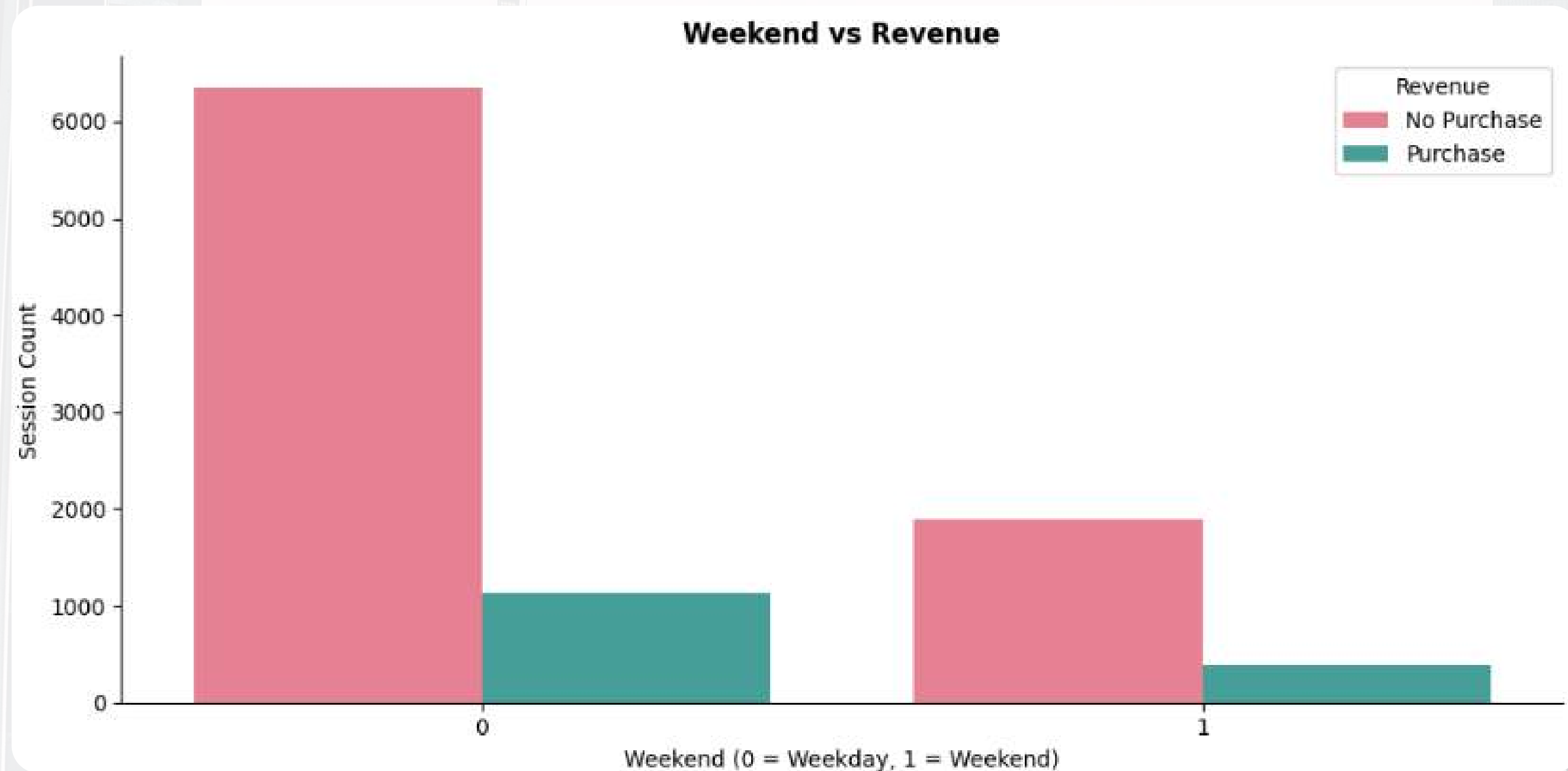
PURCHASE BEHAVIOR ACROSS VISITOR TYPES AND VISIT TIMING



Purchase Behavior Across Visitor Types

- **Returning visitors drive most purchases**, showing that prior engagement strongly increases¹⁶ conversion. This highlights **customer retention as a key revenue driver**.

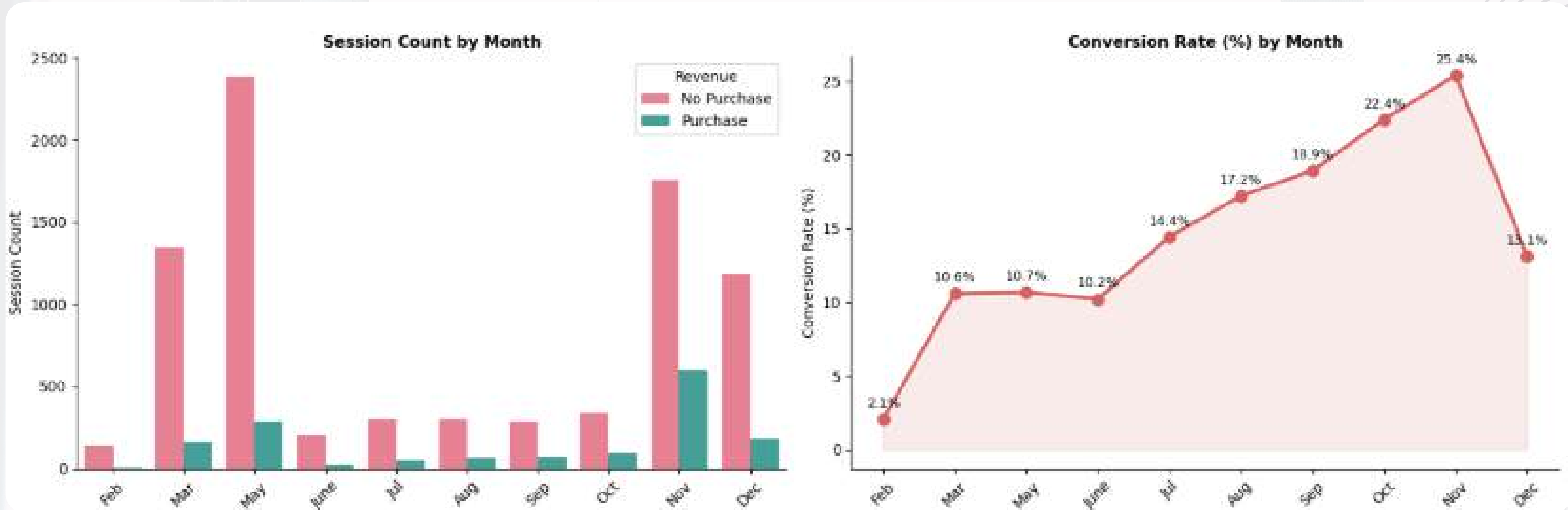
PURCHASE BEHAVIOR ACROSS VISITOR TYPES AND VISIT TIMING



Purchase Behavior Across Weekend and Weekday Sessions

- **Weekend visitors show higher purchase rates** despite lower traffic, indicating stronger buying¹⁷ intent. This suggests **visit timing adds predictive value** for identifying purchasing behavior.

NOVEMBER LEADS BOTH VOLUME AND CONVERSION QUALITY

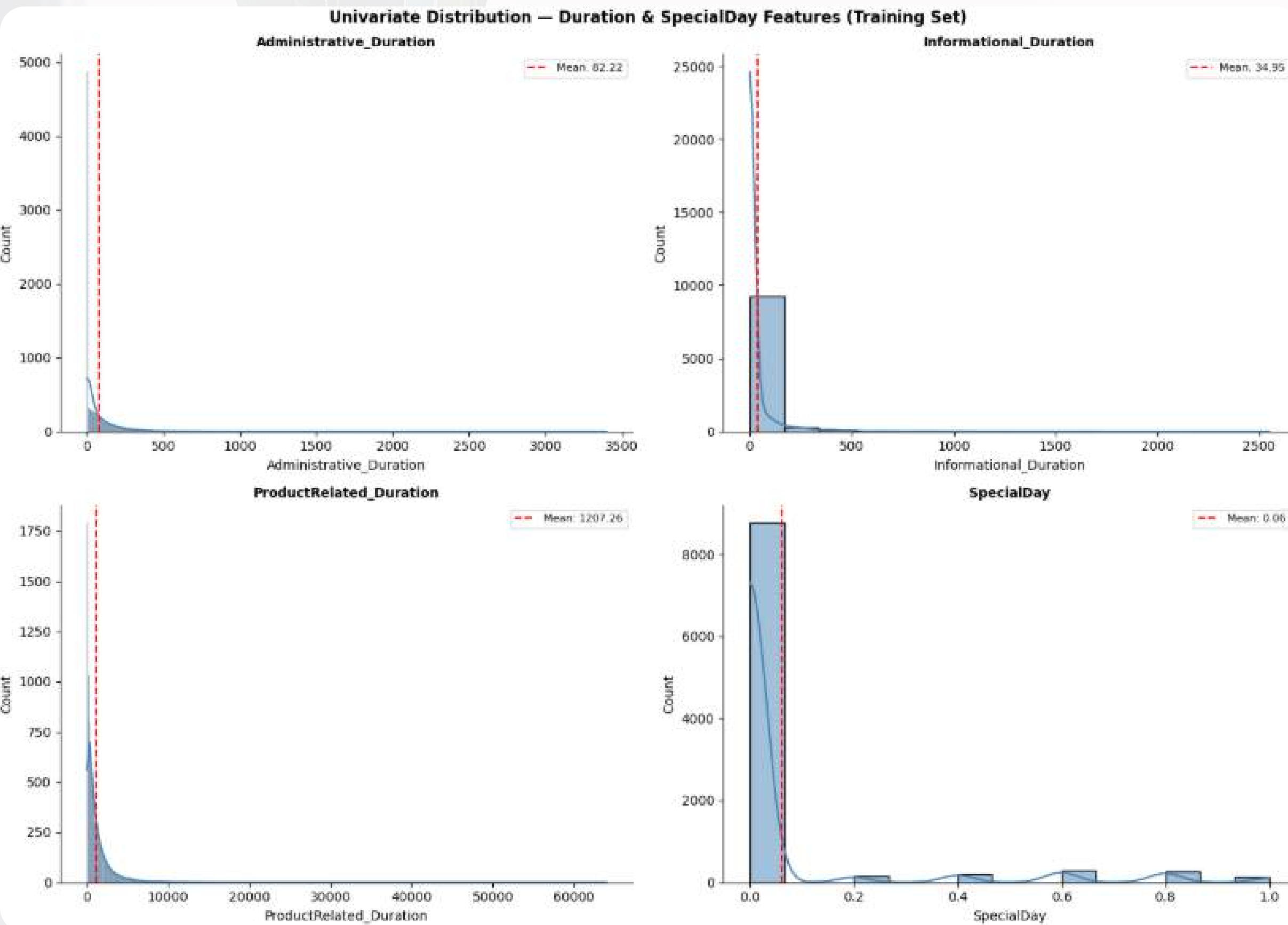


Monthly Session Volume and Conversion Rate

- November had the highest traffic and conversion rate (25.4%), showing strong seasonal effects. Conversion rose throughout the year but dropped in December despite high traffic. This suggests purchase intent peaks during major promotions, making month a useful predictor.

3.6

MOST VISITORS BROWSE CASUALLY - A FEW ENGAGE DEEPLY



Univariate Distribution of Duration and Special Day Features

- Numeric features are highly right-skewed, with most sessions low in activity and a few high-engagement outliers.
- These outliers are meaningful and retained. StandardScaler is applied for distance-based models.

4.

DATA CLEANING *and* **TRANSFORMATION**



Train/Test Split

80% training (9,764) and 20% test (2,441) using stratified sampling, preserving the 15.6% purchase rate. Test set was held out for final evaluation.

Outlier Analysis

```
=== OUTLIER ANALYSIS (IQR rule, threshold = 1.5) ===
```

Feature	Outlier count	% of train
Administrative	325	3.3%
Administrative_Duration	937	9.6%
Informational	2,107	21.6%
Informational_Duration	1,917	19.6%
ProductRelated	767	7.9%
ProductRelated_Duration	751	7.7%
BounceRates	1,127	11.5%
ExitRates	780	8.0%
PageValues	2,187	22.4%
Unique rows flagged (any column)	5,041	51.6%

Categorical Encoding

One-Hot Encoding for VisitorType, ordinal mapping for Month

Feature Scaling

After encoding, data was split into X and y. StandardScaler was fitted on training data only and applied to both sets. Scaled data was used for Logistic Regression, KNN, and MLP, while Random Forest used unscaled data.

→ **Outliers were kept** because they contain meaningful high-engagement signals, and removing them would eliminate ~52% of the data and bias the dataset.

5.

MODEL DEVELOPMENT

and

EVALUATION



Model Development

Step 1



Data Partitioning

- Dataset was split into training and test sets.
- The test set was strictly held out and not used during training or model selection to ensure unbiased evaluation.

Step 2



4 models

Four models were developed:

- Logistic Regression
- KNN
- Random Forest
- MLP

Step 3



K-Fold Cross-Validation

- 10-fold cross-validation was applied to the training set.
- F1-Macro was used as the main metric

Step 4



Model selection

- Models were compared based on cross-validation performance.

Model Development

Step 1



Data Partitioning

```
train.shape, test.shape
```

```
((9764, 18), (2441, 18))
```

- Dataset was split into **80% training** (9,764 sessions) and **20% test** (2,441 sessions).
- **Stratified sampling** was applied based on **Revenue** to preserve class distribution.
- The test set was kept completely unseen during training and model selection to **ensure unbiased evaluation**.

Model Development

Step 1



Data Partitioning

- Dataset was split into training and test sets.
- The test set was strictly held out and not used during training or model selection to ensure unbiased evaluation.

Step 2



4 models

Four models were developed:

- Logistic Regression
- KNN
- Random Forest
- MLP

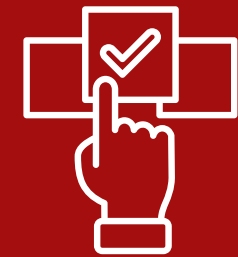
Step 3



K-Fold Cross-Validation

- 10-fold cross-validation was applied to the training set.
- F1-Macro was used as the main metric

Step 4



Model selection

- Models were compared based on cross-validation performance.

Model Development

Step 2



4 models

Four models were developed: Logistic Regression, KNN, Random Forest, and MLP.

Each model represents a different learning approach:

- **Logistic Regression** → *linear baseline*
- **KNN** → *distance-based method*
- **Random Forest** → *ensemble learning*
- **MLP** → *neural network (non-linear)*

This allows a **comprehensive comparison** across different modelling techniques.

Logistic Regression – Mean F1-Macro (10-fold CV): 0.7765 ± 0.0192

KNN (k=99) – Mean F1-Macro (10-fold CV): 0.6045 ± 0.0205

Random Forest – Mean F1-Macro (10-fold CV): 0.8048 ± 0.0121

MLP Neural Network – Mean F1-Macro (10-fold CV): 0.7669 ± 0.0178

Model Development

Step 1



Data Partitioning

- Dataset was split into training and test sets.
- The test set was strictly held out and not used during training or model selection to ensure unbiased evaluation.

Step 2



4 models

Four models were developed:

- Logistic Regression
- KNN
- Random Forest
- MLP

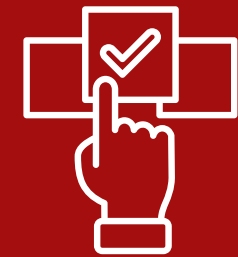
Step 3



K-Fold Cross-Validation

- 10-fold cross-validation was applied to the training set.
- F1-Macro was used as the main metric

Step 4

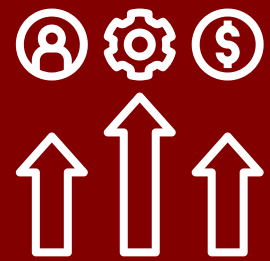


Model selection

- Models were compared based on cross-validation performance.

Model Development

Step 3



K-Fold Cross-Validation

```
=== 10-FOLD CROSS-VALIDATION RESULTS (Training Set) ===  
Logistic Regression : 0.7765 ± 0.0192  
KNN (k=99)          : 0.6045 ± 0.0205  
Random Forest       : 0.8048 ± 0.0121  
MLP Neural Network  : 0.7669 ± 0.0178
```

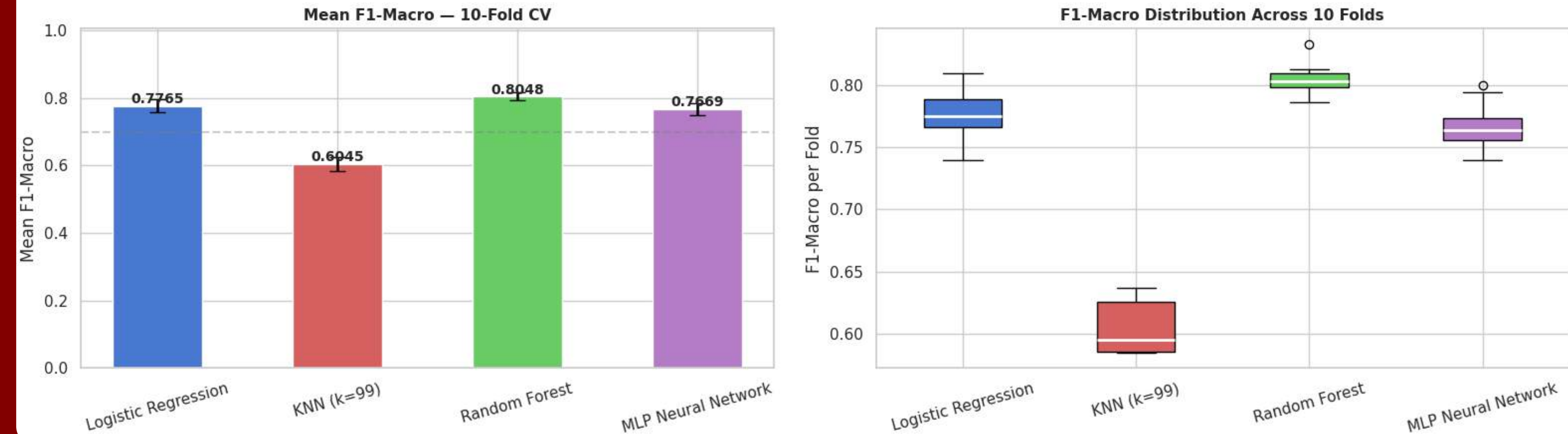
- **10-fold cross-validation** was applied **on the training set** to ensure **robust evaluation**. The data was split into 10 folds, where each fold was used once as validation while the others were used for training.
- **F1-Macro** was selected as **the main metric** due to class imbalance, ensuring equal importance for both **purchase** and **non-purchase classes**.
- **Class weighting** was applied where supported to **reduce bias** toward the majority class.

Model Development

Step 3

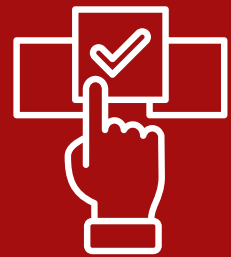
K-Fold Cross-Validation

Model Comparison — Cross-Validation on Training Set



Model Development

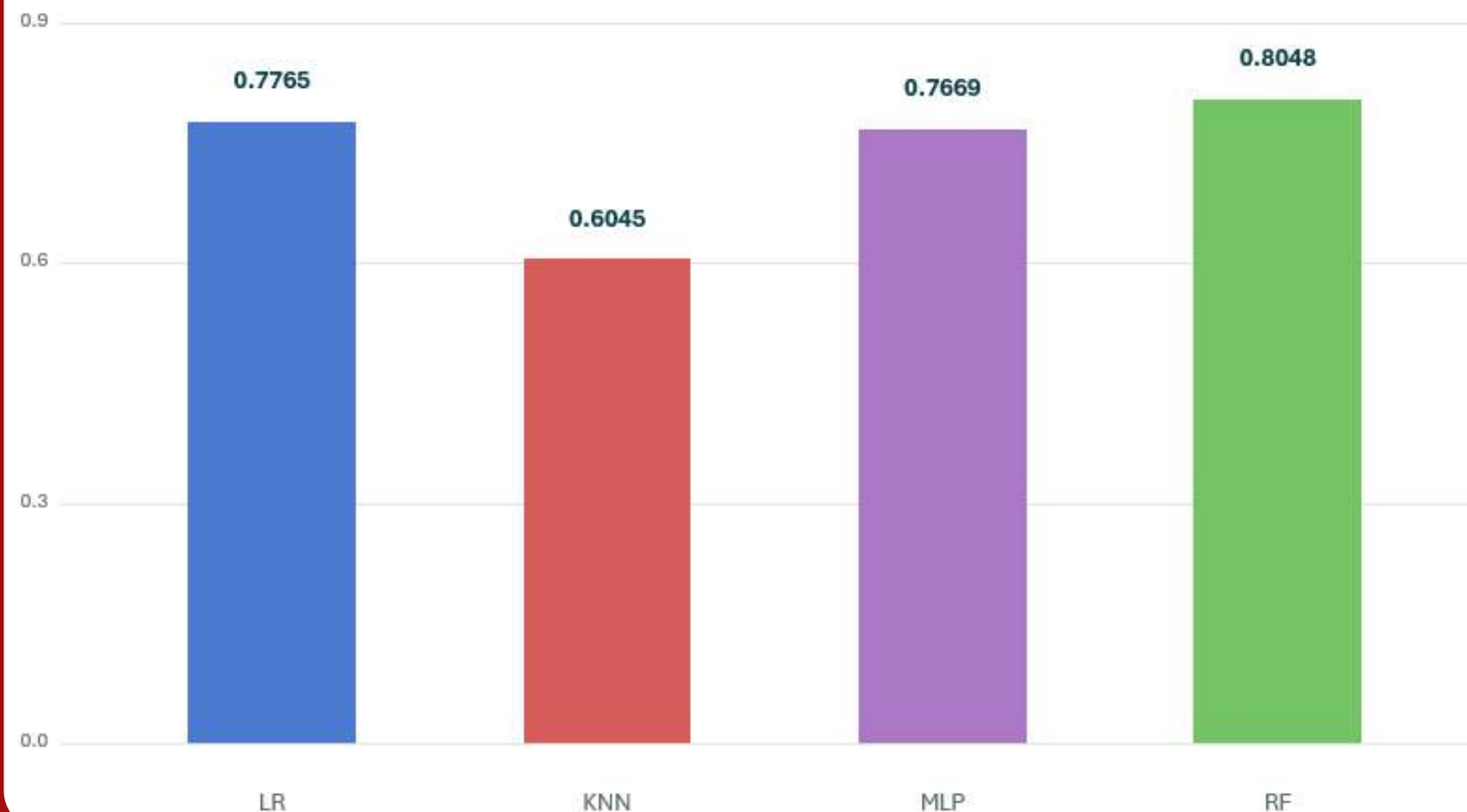
Step 4



- Models were compared **based on their mean F1-Macro and stability across folds.**
 - Random Forest achieved the highest performance and lowest variation,** indicating both **accuracy and consistency.**
- Therefore, it was selected as the final model for test evaluation.

Model selection

Mean F1-Macro — 10-Fold CV



```

=== FINAL MODEL – RANDOM FOREST – TEST SET RESULTS ===
              precision    recall  f1-score   support

         0       0.94        0.94        0.94        2059
         1       0.68        0.68        0.68         382

 accuracy                0.90        2441
 macro avg              0.81        0.81        0.81        2441
 weighted avg           0.90        0.90        0.90        2441
    
```


Model Development

Step 4: Model selection

Accuracy

90.00%

overall sessions

Precision

68.25%

purchase class

Recall

67.54%

purchase class

F1-score

67.89%

purchase class

Evaluation: Random Forest

Predicted

Non-buyer

1,939

True Negative

Buyer

120

False Positive

Non-buyer

Buyer

124

False Negative

258

True Positive

BUSINESS RISK

124 missed buyers

False negatives represent visitors who actually purchased but were not detected by the model.

Interpretation: prioritize recall carefully, but keep F1-score to control false positives and marketing cost.

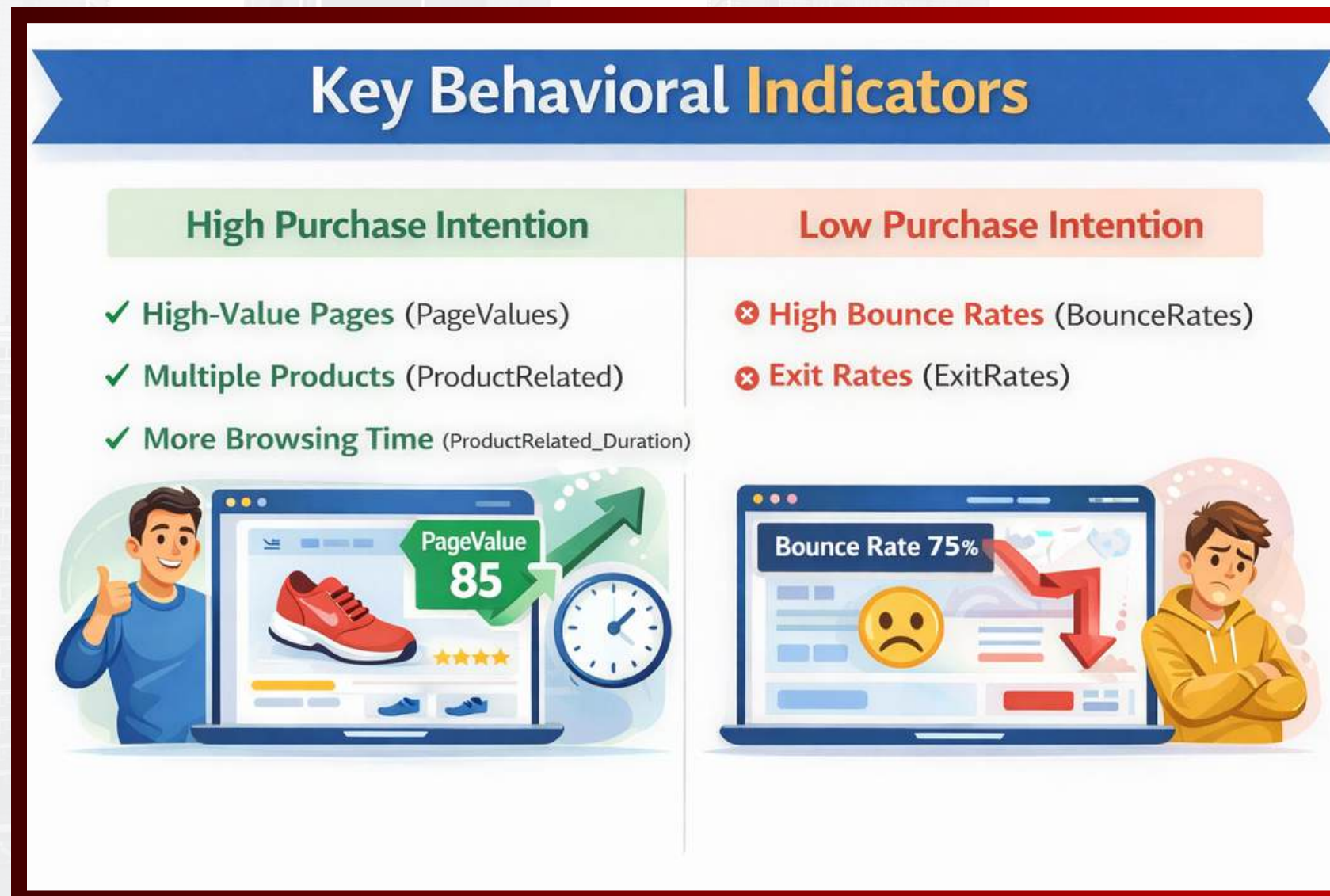
6.

INTERPRETATION & INSIGHTS



MODEL SIGNIFICANCE

Purchase-Intention Scoring Tool: Instead of treating all visitors equally, the model scores and prioritizes sessions with strong buying signals to maximize revenue.





TARGETED MARKETING STRATEGY

PERSONALIZED INTERVENTIONS

- Trigger timely actions (product recommendations, checkout reminders, limited-time offers, free shipping, or live chat) only for high-probability buyers to reduce unnecessary marketing costs.

MINIMIZE LOST OPPORTUNITIES

- Reduce false negatives (missed buyers) by adjusting the prediction threshold to increase recall, balancing the trade-off with potential increases in marketing costs.



3 PRACTICAL ACTIONS

3 Practical Actions

Prioritize High-Intent Visitors

Segment and target users based on **Model Scores** & **PageValues**.



Optimize Product Pages

Enhance descriptions, images, reviews & recommendations.



Reduce Bounce & Exit Rates

Improve landing pages, speed, & CTA placements.



THANK YOU

PRESENTED BY : POLYGON

2026

